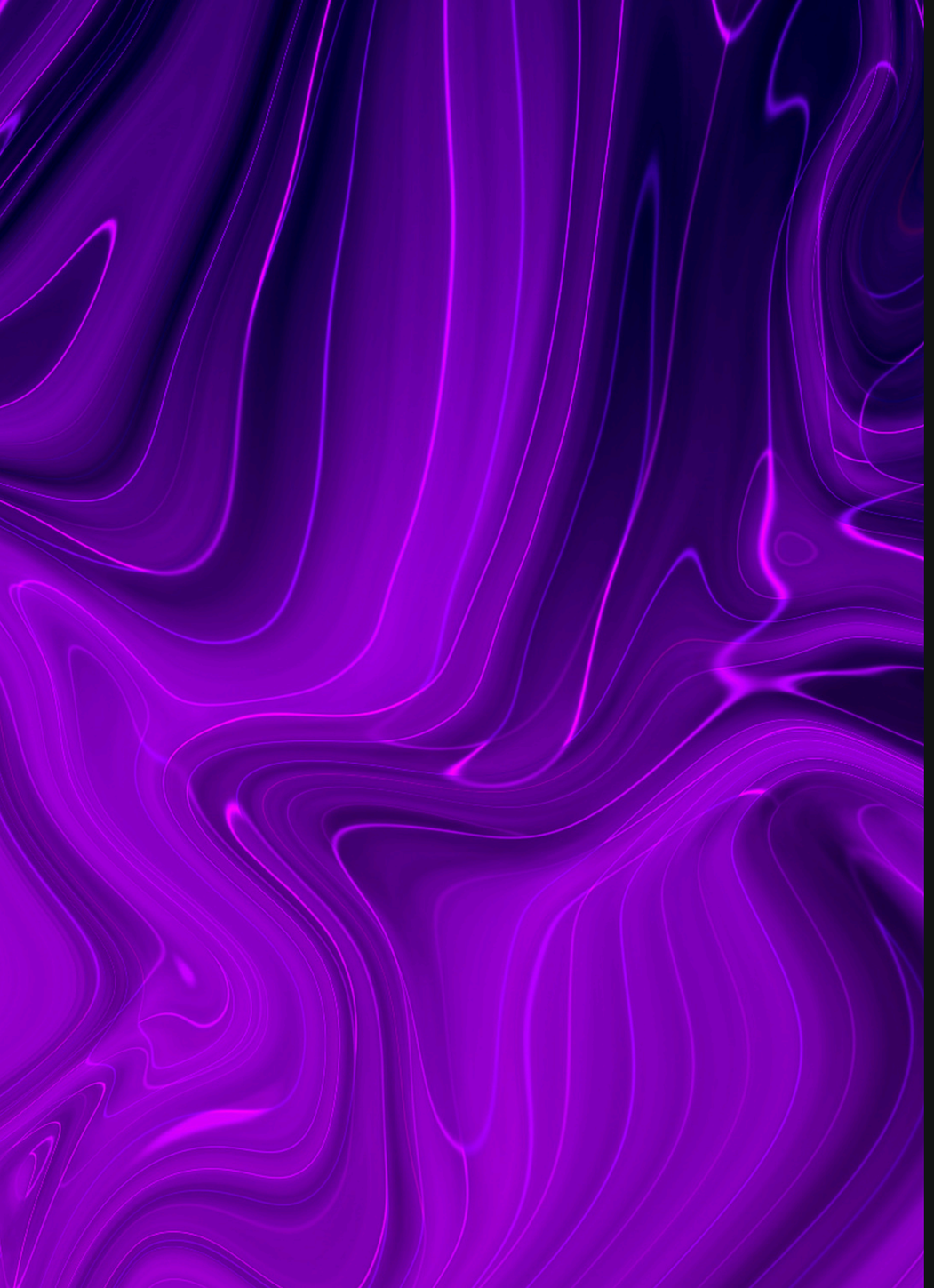
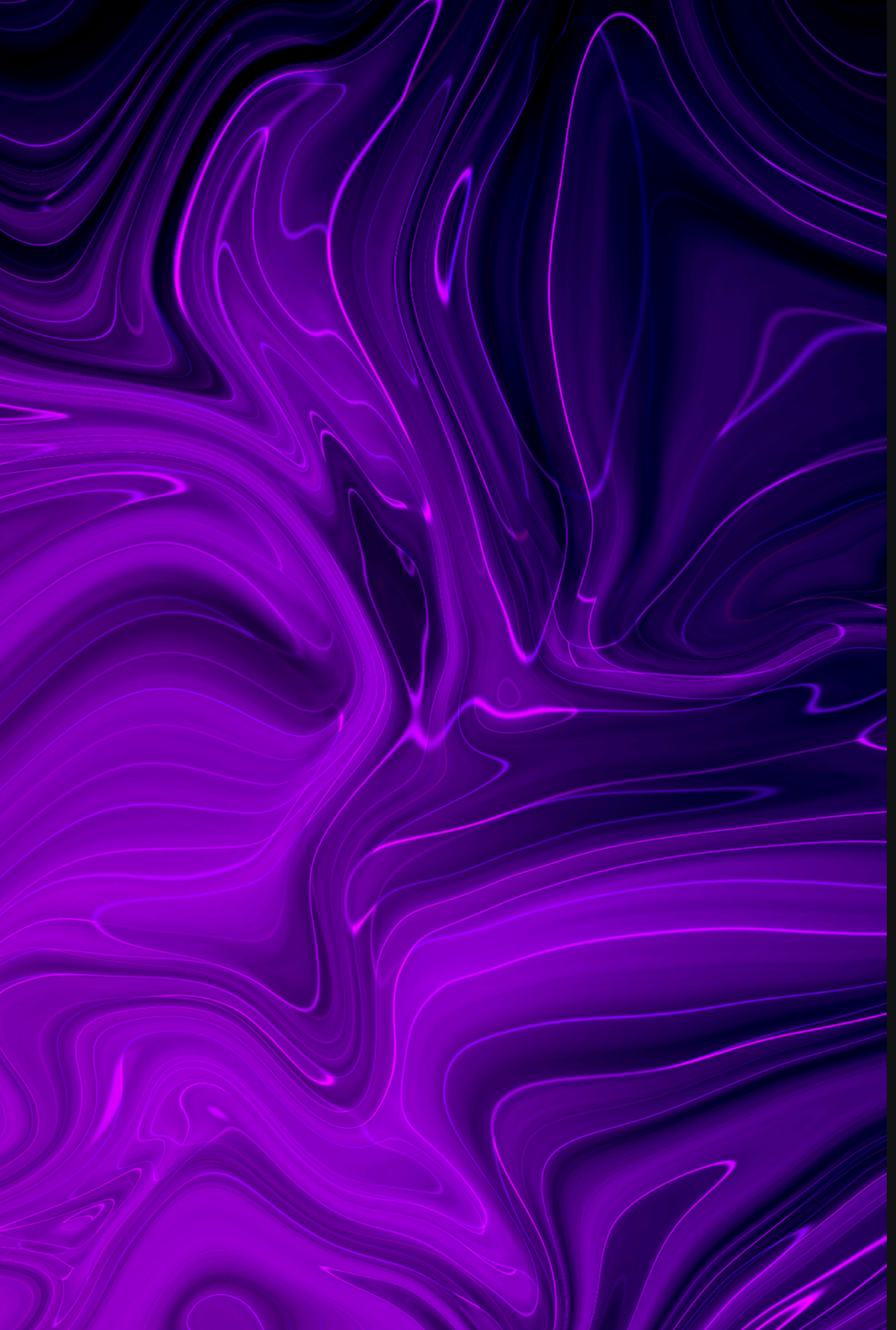


TS Academy - Cybersecurity Online Learning Platform

Bridging the cybersecurity skills gap with accessible, interactive education.



Problem Statement & Motivation

Skills Gap

Rapidly evolving digital threats

Demand

High need for cybersecurity professionals

Training Need

Accessible, practical resources required

Solution

Theory plus hands-on practice

Platform Overview - Three Core Pillars

Theory Lessons

- Structured concepts
- Foundational knowledge

Interactive Labs

- Simulated environments
- Real-world scenarios

Admin Dashboard

- Platform management
- User and content tools

Technical Architecture Overview

Frontend

React.js with TypeScript

Backend

Node.js with Express.js

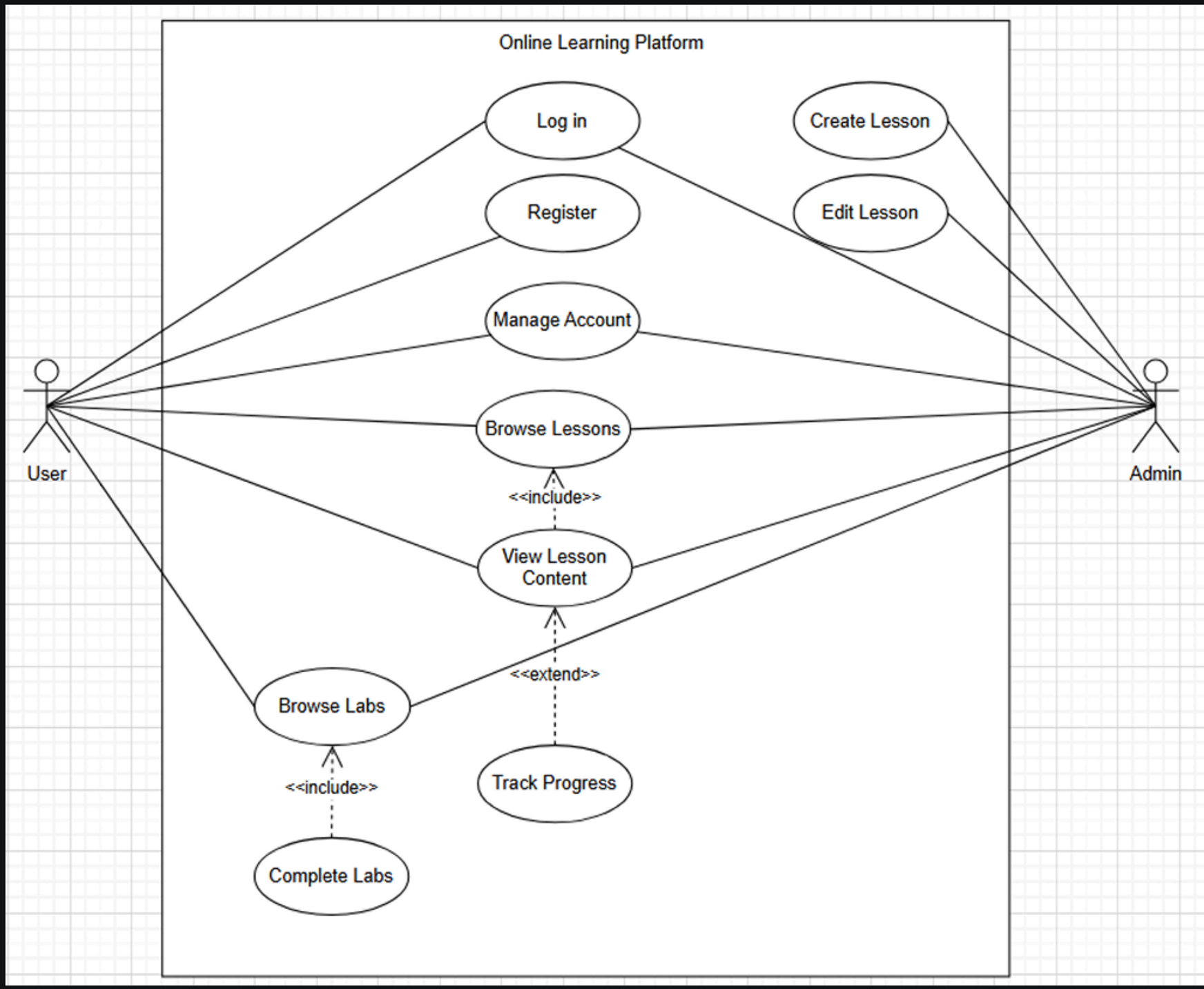
Database

MySQL for data persistence

Container Platform

Docker for labs

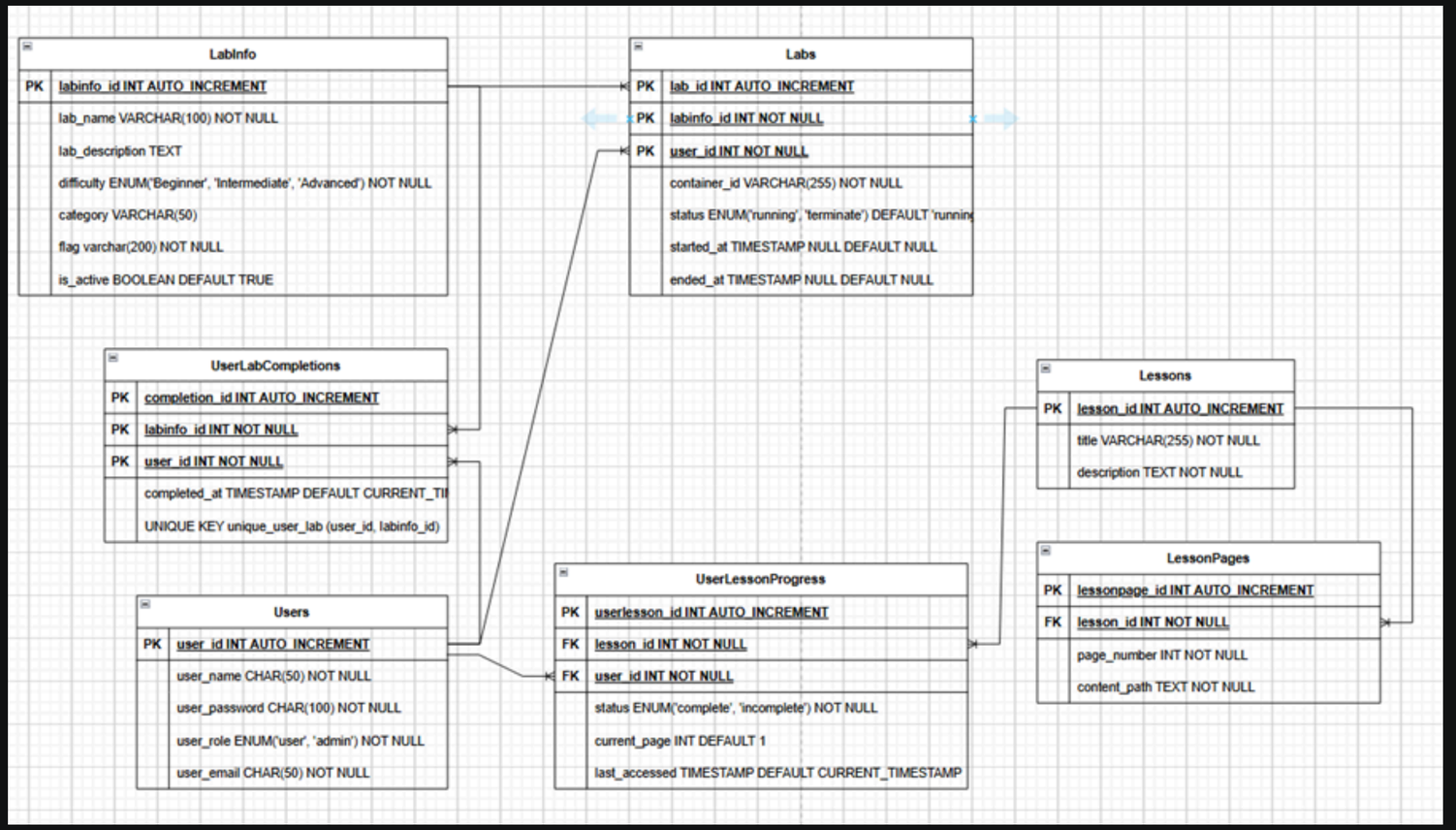
Use case diagram



Database Design

Core Entities

- Users
- Lessons & LessonPages
- Labs & LabInfo
- Progress Tracking



Frontend Architecture

Component Design

Feature-based, reusable UI components

State Management

React hooks and Context API

Routing

Protected and nested routes

Backend Architecture Deep Dive

MVC Pattern

Middleware

Authentication and JWT utilities

Security

End-to-end encryption, CORS protection

API Design & Endpoints

Authentication API

Login, register, session management

User Management API

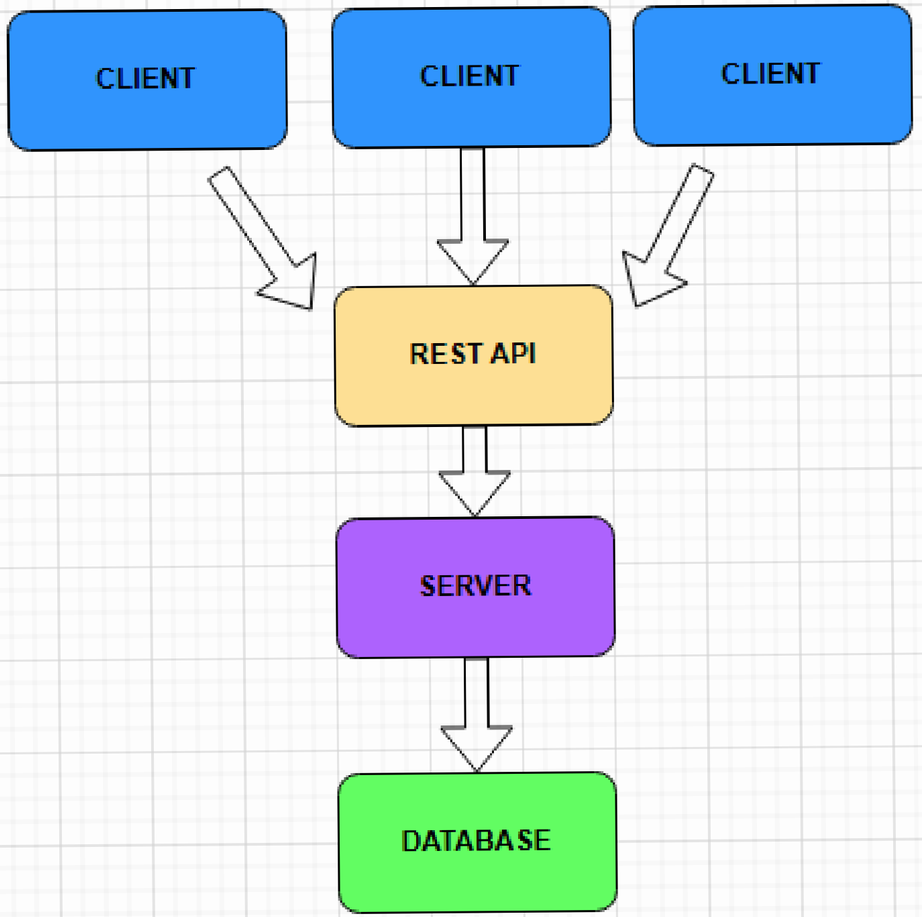
Profile and account settings

Learning Content API

CRUD for lessons and pages

Interactive Lab API

Container and flag management





Security Implementation

Authentication

JWT tokens, password hashing, session timeout

Lab Security

Docker container isolation and resource cleanup

API Security

Protected routes, input validation, sanitization

DEMO

Future Enhancements

AI-Powered Features

- Personalized feedback systems
- Intelligent learning path recommendations

Advanced Lab Environments

- SOC simulation scenarios
- Advanced penetration testing
- Forensic analysis labs

Thank You!